Introduction to Mathematics for Deep Learning: Final Solutions

1. **Mix of Questions** *(40 points, 10 points each)*

   (a) Let A be an $m \times n$ matrix ($m \geq n$), and its singular values are $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$. Show that $\min_{\|x\|_2=1} \|Ax\|_2 = \sigma_n$.

   (b) Let $A \in \mathbb{R}^{m\times k}$ and $C \in \mathbb{R}^{n\times k}$, and let $b \in \mathbb{R}^m$, $d \in \mathbb{R}^n$. For a fixed $\mu > 0$, consider the optimization problem

   $$\min_{x\in\mathbb{R}^k} \Big(\|Ax - b\|_2^2 \; + \; \mu \, \|Cx - d\|_2^2 \Big).$$

   Find the minimum–norm minimizer $x^\star$. You can define new matrices by combining A, C, b, d.

   (c) Let $\mathbb{T} : \mathbb{R}^n \to \mathbb{R}^n$ be a *nonexpansive* operator:

   $$\|\mathbb{T}(x) - \mathbb{T}(y)\|_2 \leq \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n.$$

   If z_1 and z_2 are fixed points of $\mathbb{T}$, then show that $z_\theta = \theta z_1 + (1-\theta) z_2$ is also a fixed point for any $0 < \theta < 1$.
   Hint: Consider the triangle inequality:

   $$\|z_2 - z_1\|_2 = \|\mathbb{T}(z_2) - \mathbb{T}(z_1)\|_2 \leq \|\mathbb{T}(z_2) - \mathbb{T}(z_\theta)\|_2 + \|\mathbb{T}(z_\theta) - \mathbb{T}(z_1)\|_2.$$

   (d) Let $0 < L_1 < \frac{L_2}{2}$ and consider the function

   $$f(x) = \frac{L_2}{2}\|x\|^2, \qquad x \in \mathbb{R}^n.$$

   You mistakenly believe that f is L_1-smooth and run gradient descent with stepsize $\alpha = 1/L_1$:

   $$x_{k+1} = x_k - \alpha \nabla f(x_k).$$

   Show that f is *not* L_1-smooth, and prove that the gradient descent iterates diverge (are unbounded) for every initialization $x_0 \neq 0$.

Solution.

(a) Let A be an $m \times n$ matrix with $m \geq n$, and let its singular value decomposition (SVD) be

$$A = U \Sigma V^\top,$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal, and

$$\Sigma = \begin{bmatrix} \operatorname{diag}(\sigma_1, \ldots, \sigma_n) \\ \mathbf{0} \end{bmatrix}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0.$$

For any $x \in \mathbb{R}^n$ with $\|x\|_2 = 1$, define

$$y := V^\top x.$$

Since V is orthogonal, we have $\|y\|_2 = \|x\|_2 = 1$. Then

$$Ax = U\Sigma V^\top x = U\Sigma y.$$

Because U is orthogonal, it preserves norms, so

$$\|Ax\|_2 = \|U\Sigma y\|_2 = \|\Sigma y\|_2.$$

Write $y = (y_1, \ldots, y_n)^\top$. Then

$$\|\Sigma y\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 y_i^2 \geq \sigma_n^2 \sum_{i=1}^{n} y_i^2 = \sigma_n^2 \|y\|_2^2 = \sigma_n^2.$$

Thus, for all x with $\|x\|_2 = 1$,

$$\|Ax\|_2 \geq \sigma_n.$$

To see that this lower bound is attained, take

$$y = e_n = (0, \ldots, 0, 1)^\top, \quad \text{so that} \quad x = Vy$$

and note that $\|x\|_2 = \|Vy\|_2 = \|y\|_2 = 1$. Then

$$\|Ax\|_2 = \|\Sigma y\|_2 = \|\sigma_n e_n\|_2 = \sigma_n.$$

Therefore,

$$\min_{\|x\|_2=1} \|Ax\|_2 = \sigma_n.$$

(b) Define the stacked matrix and vector

$$\widetilde{A} = \begin{bmatrix} A \\ \sqrt{\mu}\, C \end{bmatrix} \in \mathbb{R}^{(m+n)\times k}, \qquad \widetilde{b} = \begin{bmatrix} b \\ \sqrt{\mu}\, d \end{bmatrix} \in \mathbb{R}^{m+n}.$$

Then, for any $x \in \mathbb{R}^k$,

$$\|\widetilde{A}x - \widetilde{b}\|_2^2 = \|Ax - b\|_2^2 + \|\sqrt{\mu}\, Cx - \sqrt{\mu}\, d\|_2^2 = \|Ax - b\|_2^2 + \mu \, \|Cx - d\|_2^2.$$

Hence the original problem is equivalent to the standard least-squares problem

$$\min_{x \in \mathbb{R}^k} \|\widetilde{A}x - \widetilde{b}\|_2^2.$$

The set of minimizers of this least-squares problem is (possibly) an affine subspace; among all minimizers, the one with smallest Euclidean norm is given by the Moore–Penrose pseudoinverse:

$$x^\star = \widetilde{A}^\dagger \widetilde{b} = \begin{bmatrix} A \\ \sqrt{\mu}\, C \end{bmatrix}^\dagger \begin{bmatrix} b \\ \sqrt{\mu}\, d \end{bmatrix}.$$

(c) Let $z_\theta = \theta z_1 + (1-\theta) z_2$. We want to show that $\mathbb{T}(z_\theta) = z_\theta$.
Using nonexpansiveness and the fact that $\mathbb{T}(z_1) = z_1$ and $\mathbb{T}(z_2) = z_2$, we have

$$\|\mathbb{T}(z_\theta) - z_1\|_2 = \|\mathbb{T}(z_\theta) - \mathbb{T}(z_1)\|_2 \leq \|z_\theta - z_1\|_2 = (1-\theta)\|z_2 - z_1\|_2,$$

and similarly,

$$\|\mathbb{T}(z_\theta) - z_2\|_2 = \|\mathbb{T}(z_\theta) - \mathbb{T}(z_2)\|_2 \leq \|z_\theta - z_2\|_2 = \theta\|z_2 - z_1\|_2.$$

Now apply the triangle inequality:

$$\|z_2 - z_1\|_2 = \|\mathbb{T}(z_2) - \mathbb{T}(z_1)\|_2 \leq \|\mathbb{T}(z_2) - \mathbb{T}(z_\theta)\|_2 + \|\mathbb{T}(z_\theta) - \mathbb{T}(z_1)\|_2.$$

Substituting the bounds above,

$$\|z_2 - z_1\|_2 \leq \theta\|z_2 - z_1\|_2 + (1-\theta)\|z_2 - z_1\|_2 = \|z_2 - z_1\|_2.$$

Thus equality holds throughout. Equality in the triangle inequality occurs only when all points lie on the same line segment in the exact proportional positions. This forces

$$\mathbb{T}(z_\theta) = z_\theta.$$

Therefore z_θ is a fixed point of $\mathbb{T}$ for every $0 < \theta < 1$, and hence the fixed-point set of a nonexpansive map is convex.

(d) We first compute the gradient:

$$\nabla f(x) = L_2 x.$$

Hence, for any $x, y \in \mathbb{R}^n$,

$$\|\nabla f(x) - \nabla f(y)\| = \|L_2(x-y)\| = L_2\|x-y\|.$$

Therefore ∇f is Lipschitz with constant *exactly* L_2. In particular, f is L_2-smooth. Moreover, if f were L_1-smooth, we would have

$$L_2\|x-y\| = \|\nabla f(x) - \nabla f(y)\| \leq L_1\|x-y\| \quad \forall x \neq y,$$

which would imply $L_2 \leq L_1$, contradicting $L_1 < \frac{L_2}{2} < L_2$. Thus f is *not* L_1-smooth.
Next, run gradient descent with $\alpha = 1/L_1$:

$$x_{k+1} = x_k - \frac{1}{L_1}\nabla f(x_k) = x_k - \frac{1}{L_1}L_2 x_k = \Big(1 - \frac{L_2}{L_1}\Big)x_k.$$

This is a linear recursion, so

$$x_k = \Big(1 - \frac{L_2}{L_1}\Big)^k x_0.$$

Taking norms gives

$$\|x_k\| = \left|1 - \frac{L_2}{L_1}\right|^k \|x_0\|.$$

Since $L_1 < \frac{L_2}{2}$, we have $\frac{L_2}{L_1} > 2$, hence

$$\left|1 - \frac{L_2}{L_1}\right| = \frac{L_2}{L_1} - 1 > 1.$$

Therefore, if $x_0 \neq 0$ then $\|x_k\| \to \infty$ as $k \to \infty$, i.e., the iterates diverge. (If $x_0 = 0$, then $x_k \equiv 0$ for all k and the iteration stays at the minimizer.)

2. **Decorrelation Transform.** Consider a zero-mean random vector $X \in \mathbb{R}^d$ with distribution

$$X \sim \mathcal{N}(0, \Sigma),$$

where the covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric and positive definite with rank d. *(30 points)*

(a) Let $\Sigma = U \Lambda U^\top$ be an eigenvalue decomposition of Σ, where

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d).$$

Show that $\lambda_i > 0$ for all i. *(5 points)*

(b) Let $A \in \mathbb{R}^{d \times d}$ be a matrix, and define

$$Y = AX.$$

We know that Y is again a d-dimensional Gaussian random vector. Find its mean and covariance matrix. *(10 points)*

(c) Find a matrix A such that

$$Y = AX \sim \mathcal{N}(0, I_d),$$

where I_d is the $d \times d$ identity matrix. (That is, the components of Y are i.i.d. $\mathcal{N}(0, 1)$.) *(15 points)*

Solution.

(a) Since $\Sigma u_i = \lambda_i u_i$, we have $u_i^\top \Sigma u_i = \lambda_i > 0$ from positive definiteness and full-rankness.

(b) We are given

$$X \sim \mathcal{N}(0, \Sigma), \qquad Y = AX, \quad A \in \mathbb{R}^{d \times d}.$$

- *Mean of Y.*

$$\mathbb{E}[Y] = \mathbb{E}[AX] = A\,\mathbb{E}[X] = A \cdot 0 = 0.$$

- *Covariance of Y.* Since $\mathbb{E}[X] = 0$, we have

$$\mathrm{Cov}(Y) = \mathbb{E}\big[YY^\top\big] = \mathbb{E}\big[(AX)(AX)^\top\big] = \mathbb{E}\big[AXX^\top A^\top\big] = A\,\mathbb{E}[XX^\top]\,A^\top = A\Sigma A^\top.$$

Thus,

$$Y \sim \mathcal{N}\big(0,\, A\Sigma A^\top\big).$$

(c) From part (b), if $Y = AX$, then

$$Y \sim \mathcal{N}\big(0,\, A\Sigma A^\top\big).$$

We want $Y \sim \mathcal{N}(0, I_d)$, so we require

$$A\Sigma A^\top = I_d.$$

Using the decomposition from part (a), we can write

$$\Sigma = U\Lambda U^\top,$$

with U orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$.
Define

$$\Lambda^{-1/2} := \mathrm{diag}\big(\lambda_1^{-1/2}, \ldots, \lambda_d^{-1/2}\big),$$

and set

$$A := \Sigma^{-1/2} := U\Lambda^{-1/2}U^\top.$$

Then

$$A\Sigma A^\top = \big(U\Lambda^{-1/2}U^\top\big)\big(U\Lambda U^\top\big)\big(U\Lambda^{-1/2}U^\top\big)^\top.$$

Since U is orthogonal, we have $U^\top U = I_d$ and $(U\Lambda^{-1/2}U^\top)^\top = U\Lambda^{-1/2}U^\top$, so

$$A\Sigma A^\top = U\Lambda^{-1/2}\underbrace{U^\top U}_{I_d}\Lambda\underbrace{U^\top U}_{I_d}\Lambda^{-1/2}U^\top = U\big(\Lambda^{-1/2}\Lambda\Lambda^{-1/2}\big)U^\top.$$

But

$$\Lambda^{-1/2}\Lambda\Lambda^{-1/2} = \mathrm{diag}(\lambda_1^{-1/2}\lambda_1\lambda_1^{-1/2}, \ldots, \lambda_d^{-1/2}\lambda_d\lambda_d^{-1/2}) = \mathrm{diag}(1, \ldots, 1) = I_d.$$

Therefore,

$$A\Sigma A^\top = UI_dU^\top = UU^\top = I_d.$$

Hence, with

$$A = \Sigma^{-1/2} = U\Lambda^{-1/2}U^\top,$$

we have

$$Y = AX \sim \mathcal{N}(0, I_d),$$

which means the coordinates of Y are i.i.d. $\mathcal{N}(0, 1)$.

3. **Lyapunov Analysis.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be μ-strongly convex and L-smooth:

$$\mu I \preceq \nabla^2 f(x) \preceq LI, \qquad x^\star = \arg\min_x f(x).$$

Given $x_{-1}, x_0 \in \mathbb{R}^n$, consider the iteration

$$x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta(x_k - x_{k-1}), \qquad \alpha = \frac{1}{2L}, \quad \beta = \frac{1}{2}.$$

Consider the Lyapunov candidate

$$V_k := f(x_k) - f(x^\star) + c\|x_k - x_{k-1}\|^2,$$

where $c > 0$ is to be chosen. *(30 points)*

(a) Choose c so that the mixed term $\langle \nabla f(x_k), x_k - x_{k-1} \rangle$ cancels in an upper bound for $V_{k+1} - V_k$. With this c, prove the one-step estimate

$$V_{k+1} \le V_k - \frac{1}{4L}\|\nabla f(x_k)\|^2 - \frac{L}{4}\|x_k - x_{k-1}\|^2.$$

Conclude that $V_{k+1} < V_k$ whenever $x_k \neq x^\star$. *(10 points)*

(b) Let $\rho := 1 - \alpha\mu = 1 - \frac{\mu}{2L}$. Show that

$$V_{k+1} \le \rho\, V_k \qquad \text{for all } k \ge 0.$$

(10 points)

(c) Deduce exponential (geometric) convergence of x_k to $x^\star$, and give an explicit bound on $\|x_k - x^\star\|$ in terms of V_0, μ, and ρ. *(10 points)*

Solution.
Define $g_k := \nabla f(x_k)$ and $s_k := x_k - x_{k-1}$. From the iteration,

$$s_{k+1} = x_{k+1} - x_k = -\alpha g_k + \beta s_k. \tag{1}$$

We have the following

(H1) (L-smoothness) For all x, d, $\quad f(x+d) \le f(x) + \langle \nabla f(x), d \rangle + \frac{L}{2}\|d\|^2$.

(H2) (Strong convexity $\Rightarrow$ PL) For all x, $\quad \|\nabla f(x)\|^2 \ge 2\mu\big(f(x) - f(x^\star)\big)$.

(a) Apply (H1) with $x = x_k$ and $d = s_{k+1}$:

$$f(x_{k+1}) \le f(x_k) + \langle g_k, s_{k+1} \rangle + \frac{L}{2}\|s_{k+1}\|^2. \tag{2}$$

Using (1),

$$\langle g_k, s_{k+1} \rangle = \langle g_k, -\alpha g_k + \beta s_k \rangle = -\alpha\|g_k\|^2 + \beta\langle g_k, s_k \rangle,$$

and

$$\|s_{k+1}\|^2 = \| -\alpha g_k + \beta s_k\|^2 = \alpha^2\|g_k\|^2 - 2\alpha\beta\langle g_k, s_k\rangle + \beta^2\|s_k\|^2.$$

Combine these with (2) and add $c(\|s_{k+1}\|^2 - \|s_k\|^2)$ to obtain

$$V_{k+1} - V_k \le -A\|g_k\|^2 + B\langle g_k, s_k\rangle + D\|s_k\|^2,$$

where

$$A = \alpha - \frac{L\alpha^2}{2} - c\alpha^2, \qquad B = \beta(1 - L\alpha) - 2c\alpha\beta, \qquad D = \frac{L\beta^2}{2} + c(\beta^2 - 1).$$

Choose c so that $B = 0$, i.e.

$$c = \frac{1 - L\alpha}{2\alpha}.$$

With $\alpha = \frac{1}{2L}$ this gives $c = \frac{L}{2}$. For this choice (a short simplification) yields $A = \frac{\alpha}{2}$ and

$$D = \frac{L}{2} - \frac{1 - \beta^2}{2\alpha}.$$

With $\beta = \frac{1}{2}$ and $\alpha = \frac{1}{2L}$,

$$A = \frac{1}{4L}, \qquad D = \frac{L}{2} - \frac{3/4}{1/L} = \frac{L}{2} - \frac{3L}{4} = -\frac{L}{4}.$$

Hence

$$V_{k+1} - V_k \le -\frac{1}{4L}\|g_k\|^2 - \frac{L}{4}\|s_k\|^2,$$

which is the desired estimate. If $x_k \neq x^\star$, then $g_k \neq 0$ (uniqueness of minimizer), so the right-hand side is strictly negative and $V_{k+1} < V_k$.

(b) From part (a) and (H2),

$$-\frac{1}{4L}\|g_k\|^2 \le -\frac{1}{4L}\cdot 2\mu\big(f(x_k) - f(x^\star)\big) = -\frac{\mu}{2L}\big(f(x_k) - f(x^\star)\big).$$

Also $L \ge \mu$ implies

$$-\frac{L}{4}\|s_k\|^2 \le -\frac{\mu}{4}\|s_k\|^2 = -\frac{\mu}{2L}\cdot\frac{L}{2}\|s_k\|^2 = -\frac{\mu}{2L}\,c\|s_k\|^2,$$

since $c = \frac{L}{2}$. Therefore,

$$V_{k+1} \le V_k - \frac{\mu}{2L}\big(f(x_k) - f(x^\star)\big) - \frac{\mu}{2L}c\|s_k\|^2 = \Big(1 - \frac{\mu}{2L}\Big)V_k = \rho\, V_k,$$

where $\rho = 1 - \alpha\mu = 1 - \frac{\mu}{2L} \in (0, 1)$.

(c) Iterating (b) gives $V_k \le \rho^k V_0$. Since $V_k \ge f(x_k) - f(x^\star)$ and by strong convexity $f(x_k) - f(x^\star) \ge \frac{\mu}{2}\|x_k - x^\star\|^2$, we obtain

$$\|x_k - x^\star\|^2 \le \frac{2}{\mu}\,V_k \le \frac{2}{\mu}\,\rho^k V_0,$$

hence

$$\|x_k - x^\star\| \le \sqrt{\frac{2V_0}{\mu}}\;\rho^{k/2}.$$

This is geometric convergence to $x^\star$.